



CL2005

Database Systems Lab

**Animal Hospital and Shelter
Management System**

3. Paw Track

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Submitted to: Mr Ahsan Shakeel

Submitted by:

CY-A

Eman Akbar

22i-1588

Sarita Sangrez

23i-2088

Amna Ali

23i-2067

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1. Executive Summary

This project proposes an easy-to-use database system to manage an animal hospital and shelter. Currently, all processes like intake, medical records, room allocation, adoptions, and volunteer scheduling are handled manually using paper or spreadsheets, which causes delays, duplication, and errors. The new system will centralize these operations, allowing staff to intake animals, vets to record diagnoses and treatments, and admins to track adoptions, room capacity, and volunteer hours. It will reduce errors, improve efficiency, and ensure secure, reliable, and fast access to data.

2. Context & Stakeholders

2.1 Domain Background

Currently, animal shelter and hospital activities are based on the use of manual record keeping and fragmented documentation. When a new animal is admitted, the staff completes a paper intake form with information such as species, breed, age and medical status. These forms are kept physically or simple spread sheets on personal computers. Room assignments are documented on a white board or a handwritten log that can be confusing in cases when animals are changed or rooms are not available.



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Medical histories are kept by veterinarians in paper sheets which have been stapled in the file of each animal. All treatment, diagnosis, and prescriptions are handwritten and hence difficult to find or share instantly with other personnel. Scheduling is managed through wall planners or through messaging applications, and therefore often results in duplications or omissions.

Adoptions are processed on application forms, which are normally filled manually and the documents are stored in filing cabinets. Tracking is sometimes done via spread sheets, which are not consistent and are kept in different locations by various members of staff. The volunteers are booked by phone calls, WhatsApp group or even sticky notes and there is no trusted central account of their shifts, hours or duties.

The shelter usually deals with:

- Animal Consumption: A number of animals a week, which results in dozens of active records at once.
- Medical Records: When it comes to medical records, having several treatments/checkups in a day, it becomes hundreds of records fast.
- Visits: Every day vet visit and checkups.
- Adoptions: There are several cases weekly, and these need forms and follow-up.
- Volunteer shifts: Two shifts of volunteers of the day.

Due to this disjointed and paper-based system, employees lose time in scrolling through papers, there are mistakes (duplicate bookings or misplaced medical records), and decision-making is postponed because of the absence of jeopardy summations or collated findings.

2.2 As-Is Workflow

At the moment, everything in the shelter and hospital is done manually, and records are not integrated into each other:

1. Animal Intake

- Each time a new animal is introduced the staff completes a paper form, intake form, with the particulars about the animal such as name, species, breed, age, health condition.
- It is placed in a physical file folder and the animal is manually placed in a room with the help of a white board or notebook.

2. Room Assignment



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- Employees check in rooms by viewing the white board or walking to shelter.
- When a room seems occupied, animals can be re-allocated inaccurately because there is no proper monitoring of them.

3. Medical Records

- Veterinarians write down the diagnosis, the prescription and treatment notes on paper slips or separate medical sheets stored in the file of the animal.
- It cannot be easily searched in the medical history of an animal; employees have to sift through physical folders.

4. Appointments

- The appointments are made on a wall calendar or sometimes on a shared spreadsheet.
- The problems of double-booking or miss entries occur due to the lack of automated checks.

5. Adoptions

- The interested adopters complete the application forms on paper.
- The forms are read by the admin, stored in a filing cabinet and the status of the adoption of an animal on the paper records is updated.

6. Volunteer Scheduling

- Volunteer shifts and duties are organized via telephone or WhatsApp group or sticky.
- No volunteer work or duties are recorded.

2.3 Stakeholders

Stakeholder	Responsibilities	Needs / Pain Points
Admin	Supervises the work of general activity, controls user accounts, activates / disables adoptions, traces reports.	Need integrated reports, secure access control, and audit logs. Frustrated by scattered and excessive paper files.
Veterinarian	Checkups, diagnoses, records treatment and prescriptions, schedules.	Wants to have easy access to the medical history of every animal, easy scheduling of appointments, and less paper work.



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Staff (Clerk/Attendant)	Accepts new animals, allocates rooms, updates on the animal status, appointments.	Needs to be able to easily track room capacities, search faster, and is subject to validation to prevent the entry of the same information twice.
Volunteer	Assists with animal care (feeding, cleaning, enrichment), staff and vet support.	Should have a shift/task schedule, and a record of the hours worked. Appropriately takes assignments at this time.
Adopter	Visits shelter, makes an application to adopt animals, submits papers and goes through adoption procedure process.	Requires knowledge of adoption condition, effective communication and expedited adoption.

3. Problem Statement & Objectives

Problem Statement:

Animal hospital and shelter operations are currently handled in an ineffective manner with limited paper-based forms and informal systems of tracking like registers, spread sheets, and calendars. Animal feeding, medical history, assigning rooms, adoptions and volunteer schedules are kept separately in little or no integration. This creates a number of problems: the same records or absence of them, it is hard to find out room capacities, there is a delay in finding medical histories, and there are discrepancies in documentation of adoptions. Besides, this is because the absence of centralized data renders timely production of reports, accountability, and administration of security and privacy of sensitive information, difficult on the part of the administrators. Since the shelter is working with tens of animals, constant treatments, daily visits, and regular adoptions, these manual practices are inefficient, prone to errors and unsustainable. It requires a digital, centralized database system, to enhance accuracy, minimize errors, streamline workflows, and to secure and efficiently undertake the operations of all stakeholders.

Objectives:

- Accuracy: Eliminate at least 90 percent of the duplicate or invalid records with the help of validation rules and unique ID.
- Efficiency: Uncover lookups on the animal record in less than 1 second on a dataset of 5,000 rows or less.
- Capacity Management: Automatic prevention of over-assignment of rooms and real time occupancy tracking.
- Medical Records: Keep a full treatment history on each animal, the diagnoses, prescriptions and even follow-ups.



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- Reporting: Summaries (appointments, occupancy, adoptions): Produce reports with fewer than 10 seconds per record with up to 10,000 records.
 - Security: Implement the role-based access control, authentication and audit trail of sensitive operations.
 - Data Reliability: Back up of data daily with a recovery point objective (RPO) of 1 hour and recovery time objective (RTO) of 4 hours.
 - User Experience: It should have simple and intuitive staff, veterinarian, and admin forms to reduce the amount of time and errors in input.
-

4. Requirements

4.1 Functional Requirements (FRs)

FR-1: The system shall allow intake of new animals by staff with details (unique animal ID, name, species, status, breed, date, age, shelter room)

FR-2: The system shall assign each animal to a shelter room and update/manage the rooms capacity.

FR-3: The system shall allow staff to search and view animal details by Animal ID, name, or status.

FR-4: The system shall prevent assigning an animal to a room if the capacity is full.

FR-5: The system shall allow veterinarians to record diagnosis, medication and treatments with updated next checkup date.

FR-6: The system shall link treatments to both animal and veterinarian records.

FR-7: The system shall allow staff to update appointment status (scheduling, updating and cancelling appointments) with date, assigned staff, purpose and link to animal ID.

FR-8: The system shall record adoption details (adopter details, adoption date and adoption file) linking adopter and animal changing animal status to adopted.

FR-9: The system shall ensure each adoption is linked to an adopter and an animal.

FR-10: The system shall allow assigning volunteers to tasks and shifts.



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FR-11: The system shall allow role-based control only by admin (roles: Staff, Volunteer, Veterinarian) maintaining staff records and veterinarian details.

4.2 Non-Functional Requirements (NFRs)

NFR-1: Performance: 95th-percentile latency for lookup operations shall be < 1 second for datasets < 5k rows, aggregations shall complete within 10 seconds for 10000 rows.

NFR-2: Reliability and Availability: The system shall be active during business hours (8:00 am - 10:00 pm) and data must be backed up daily to prevent loss. Acceptable RTO (Recovery Time Objective) is 4 hours and RPO (Recovery Point Objective) is 1 hour for transactional data.

NFR-3: Security and Privacy: All user authentication shall use password and role-based access (vet, staff, volunteer, admin) should be enabled. Sensitive data (owner contact, donor PII) must be stored encrypted at rest. Audit logs cannot be changed.

NFR-4: Data Quality: required fields shall be enabled (like Ids, dates etc.). Data types should be followed.

NFR-5: Usability: User-friendly forms and dashboards for easy data entry.

NFR-6: Compliance/Retention: Medical records and adoption documents shall be retained for at least 5 years.

4.3 Use Cases

Use Case ID	Title	Primary Actor	Pre-conditions	Main Flow (Steps)	Post-conditions	Exceptions
UC-1	Intake animals	Staff	Staff are logged in. Shelter room available.	1. Staff select 'Intake Animal'. 2. Enter animal details (species, breed, age, status). 3. Assign shelter room. 4. Save record.	New animal record created. Shelter room updated.	Room full → system shows 'No capacity'. Missing required info → error message.
UC-2	Schedule appointment	Staff	Staff logged in. Animal record exists.	1. Staff selects 'Schedule Appointment'. 2. Enter appointment date, time, vet, purpose.	Appointment booked and linked to animal and veterinarian.	Time conflict → system rejects booking. Invalid animal ID → error.

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				3. System checks vet availability. 4. Save appointment.		
UC-3	Record treatment	Veterinarian	Appointment completed vet logged in	1. Vet opens animal's record. 2. Add diagnosis, treatment, medication. 3. Save entry.	Treatment record saved. Follow-up date set.	Missing mandatory fields → error. Unauthorized user → access denied.
UC-4	Manage adoption	Adopter	Animal eligible for adoption. Adopter registered	1. Adopter submits adoption request. 2. Admin reviews details. 3. Collect adoption file and fee. 4. Approve adoption.	Animal status updated to 'Adopted'. Adoption record created.	Adoption rejected → update application as 'Declined'.
UC-5	Assign volunteer	Admin	Volunteer registered Admin logged in	1. Admin selects 'Assign Volunteer'. 2. Enter shift/time and role. 3. Save role.	Volunteer assigned to shift. Schedule updated.	Volunteer unavailable → system shows conflict.
UC-6	Manage vet/staff	Admin	Admin logged in	1. Admin selects 'Manage Staff'. 2. Add, update, or remove staff info. 3. Save changes.	Staff/vet records updated.	Invalid data → error message. Unauthorized → access denied.

Stakeholders → Needs → Trace to FR

Stakeholder	Needs / Pain Points	Mapped FRs
Admin	Needs reports, secure access control, and audit logs.	FR-7, FR-11
Veterinarian	Wants easy access to medical history, record	FR-5, FR-6



	treatments, and schedule checkups.	
Staff	Needs to intake animals, manage rooms, search animals, and schedule appointments.	FR-1, FR-2, FR-3, FR- 4, FR-7
Volunteer	Should have shift/task assignments and record of hours worked.	FR-10
Adopter	Requires adoption process tracking, adoption records, and updated animal status.	FR-8, FR-9

5. Data Discovery & Glossary

Entity Candidates:

- **Animal**
- **Adopter**
- **Adoption** (weak entity)
- **Staff** (generalized): {Admin, Veterinarian, Volunteer}
- **Treatment**
- **Appointment**
- **Shelter Room**

Attribute Candidates

(with data types, units, and constraints)



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Animal

- Animal ID (PK, INT)
- Name (VARCHAR, 50)
- Species (ENUM: Dog, Cat, Rabbit, etc.)
- Breed (VARCHAR, 50)
- Gender (ENUM: Male, Female)
- DateOfBirth (DATE)
- Status (ENUM: Available, InTreatment, Adopted, Deceased)
- ShelterRoomID (FK, INT)
- Age (INT, derived from DateOfBirth, in years)

Adopter

- AdopterID (PK, INT)
- FullName (VARCHAR, 100)
- Phone (VARCHAR, 15)
- Email (VARCHAR, 100, unique)
- Address (VARCHAR, 200)
- CNIC (VARCHAR, 15)

Adoption (weak entity)

- AdoptionID (PK, INT, partial key with AdopterID)
- AdoptionDate (DATE)
- AnimalID (FK, INT)
- AdopterID (FK, INT)
- AdoptionFile (VARCHAR, 255)

Staff (generalized)

- StaffID (PK, INT)
- Name (VARCHAR, 100)
- Phone (VARCHAR, 15)
- Email (VARCHAR, 100, unique)
- Role (ENUM: Admin, Veterinarian, Volunteer)

Admin (ISA Staff)

- AdminID (PK, INT, FK from StaffID)
- Salary (DECIMAL(10,2))



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Veterinarian (ISA Staff)

- VetID (PK, INT, FK from StaffID)
- Specialization (VARCHAR, 100)
- LicenseNo (VARCHAR, 20)
- Salary (DECIMAL(10,2))

Volunteer (ISA Staff)

- VolunteerID (PK, INT, FK from StaffID)
- Availability (VARCHAR, 50)

Treatment

- TreatmentID (PK, INT)
- AnimalID (FK, INT)
- VetID (FK, INT)
- TreatmentDate (DATE)
- Diagnosis (VARCHAR, 200)
- Prescription (VARCHAR, 200)

Appointment

- AppointmentID (PK, INT)
- AnimalID (FK, INT)
- VetID (FK, INT)
- DateTime (DATETIME)
- Status (ENUM: Scheduled, Completed, Cancelled)

ShelterRoom

- RoomID (PK, INT)
- RoomType (ENUM: Kennel, CatRoom, Isolation, etc.)
- Capacity (INT)
- OccupiedCount (INT, derived)

VolunteerAnimal (junction)

- VolunteerID (FK, INT)
- AnimalID (FK, INT)
- AssignedDate (DATE)
- HoursPerWeek (INT)



Business Rules (Cardinality, Optionality, Constraints)

1. Each Animal must be assigned to exactly one Shelter Room.
2. An Adopter may adopt many Animals, but each Animal can be adopted only once.
3. Adoption is dependent on Adopter - Adoption cannot exist without Adopter.
4. Each Treatment must be carried out by one Veterinarian and for one Animal.
5. A Volunteer can care for many Animals, and an Animal can be cared for by many Volunteers.
6. OccupiedCount of ShelterRoom is derived from number of assigned Animals.
7. Every Staff member belongs to exactly one subclass: Admin, Veterinarian, or Volunteer (disjoint ISA).
8. Only Admins and Veterinarians have salaries; Volunteers do not.
9. Age of Animal is derived from DateOfBirth.
10. The relationship *cares_for* between Volunteer and Animal has an attribute “Duration”, which records how long the volunteer is assigned to that animal (e.g., in weeks or months).

Assumptions for Modelling choice

- Each Animal must always be assigned to exactly one ShelterRoom while in the shelter. A room can house multiple animals up to its capacity.
- Adoption is a weak entity that cannot exist without both an Adopter and an Animal. An animal can be adopted only once.
- Only staff members classified as Veterinarians can be linked to Treatments and Appointments.
- All Admins, Veterinarians, and Volunteers are modeled as subclasses of Staff (disjoint, total specialization). Each Staff member belongs to exactly one subclass.
- Appointments are optional for Animals (not every animal has one), but if scheduled, they must involve exactly one Animal and one Veterinarian.
- The attribute *AdoptionFile* in Adoption stores only the path/URL to the digital adoption document (not the file itself).
- Only Admin and Veterinarians are salaried staff. Salary for Volunteer is NULL.

Glossary of Domain Terms

- **Adopter** – A person who legally adopts an animal from the shelter.

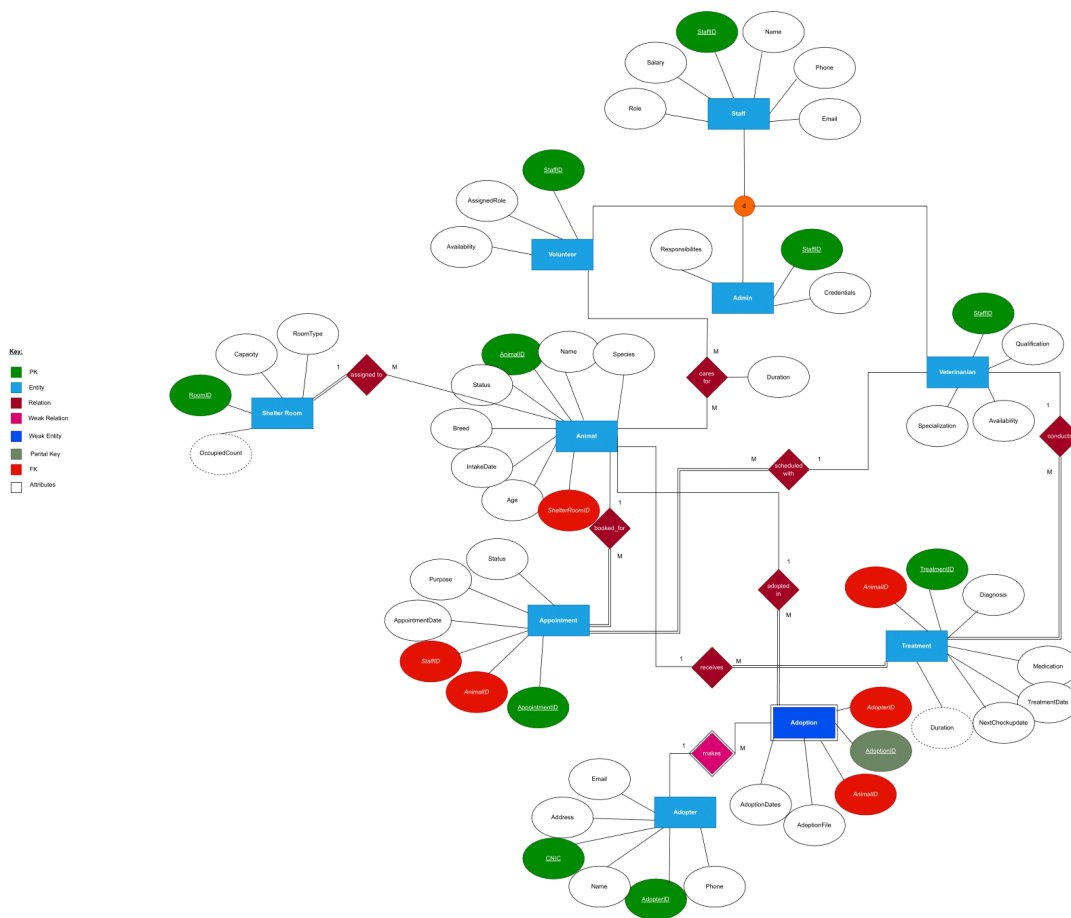


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- **Adoption** – A process recording the transfer of ownership of an animal to an Adopter.
 - **Animal** – Any creature housed in the shelter (e.g. dog, cat, rabbit).
 - **Appointment** – A scheduled meeting between an Animal and a Veterinarian.
 - **ShelterRoom** – A designated room in the shelter where Animals are kept.
 - **Staff** – General term for human resources in the system (Admin, Veterinarian, Volunteer).
 - **Veterinarian (Vet)** – A licensed staff member who diagnoses and treats Animals.
 - **Volunteer** – A non-salaried staff member who helps care for Animals.
 - **Admin** – A salaried staff member managing operations and records.
 - **Treatment** – A medical record describing a diagnosis and prescription for an Animal.
-

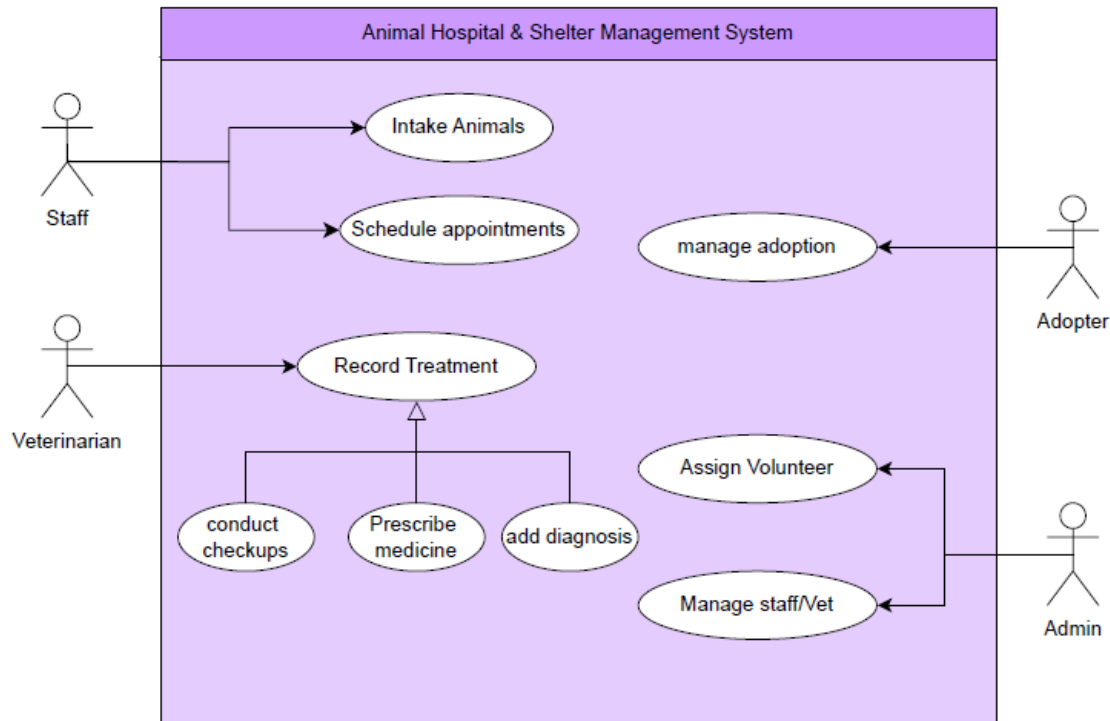


6. ER/EER Diagram





UML Use Case Diagram:



7. Risks & Assumptions

Risks:

- Missing stakeholder contribution: Staff, vets or volunteers are not always available to be interviewed and thus requirements are sometimes missing.
- Lack of records (or their inconsistency): Due to the nature of paper-based intake and treatment forms, certain errors can occur, and it is difficult to trace precise workflows.
- Adjusting regulations: Adoption policy, room capacity, or volunteer assignment policy may be adjusted during the project, and would need re-design.
- Technical limitations: The lack of infrastructure (old computers, bad internet) may slow the adoption of the system.
- Data migration challenges: When existing paper/spreadsheet data is converted to the new system, it is likely to create errors or omissions.



Assumptions:

- All members of staff pertain to only one role (Admin, Veterinarian, Volunteer).
- Each animal is occupied by a single shelter room only.
- Every adoption is a one-time thing - an animal can never be adopted again.
- Age of animals may be obtained by using Date of Birth or may be estimated at intake.
- The adopters give valid and confirmed contact details (e.g., CNIC, phone number).
- It will be working during regular hours on the shelters (08:00-22:00). Backups will be made daily according to the policy to recover the data.

8. References & Ethics

References:

- Elmasri, R., & Navathe, S. B. (2016). *Fundamentals of database systems* (7th ed.). Pearson.
- Malik, A. S. (2025). *CS2005 Database Systems (Lab) – Course lectures and project guidelines*. FAST-NUCES.
- American Society for the Prevention of Cruelty to Animals (ASPCA). (n.d.). *Shelter operations manual*. ASPCA. <https://www.asPCA.org>
- Humane Society of the United States (HSUS). (n.d.). *Standards of care in animal shelters*. HSUS. <https://www.humanesociety.org>
- World Small Animal Veterinary Association (WSAVA). (n.d.). *Guidelines for veterinary care*. WSAVA. <https://wsava.org>

Ethics:

- All the examples are anonymized in order not to reveal sensitive adopter, staff, or animal information.
- Places of personal identifiers, including CNIC or phone numbers, are filled with placeholders.
- Any observations or interviews (in case of such) were done consensually.
- Information gathered will be of academic use only and will be kept safely.



Appendices

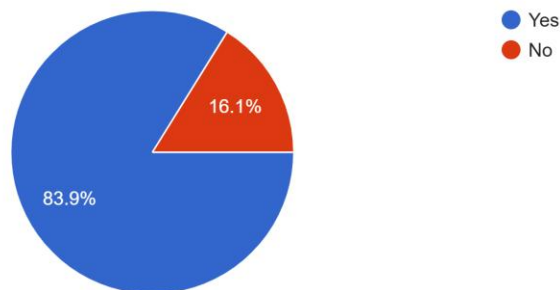
- A. Interview notes

To find the viewpoints of people around us about the pet app we circulated a google form:

And we received 31 responses, here are the results from the set of questions.

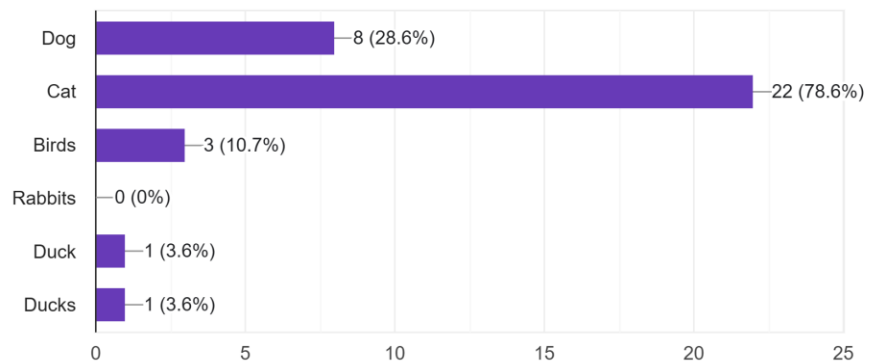
Do you currently own a pet?

31 responses



If yes, what kind of pet(s) do you own?

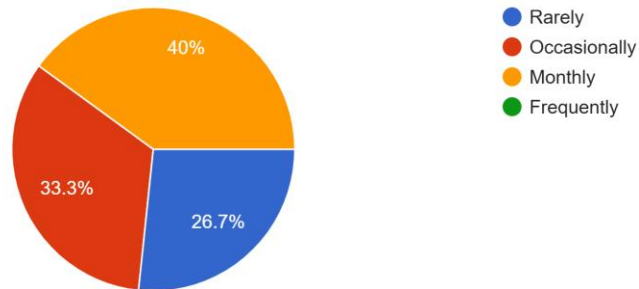
28 responses





How often do you visit a vet or pet hospital?

30 responses



What is the biggest challenge you face in managing your pet's health or care?

27 responses

Keeping track of medical records and vaccination history, scheduling vet appointments, and finding reliable pet care services and the best food appropriate for my cat.

Access to vet

Identifying the symptoms

evil vets trying to siphon money

Nothing.

Taking care when they go out

keeping up with vet visits and noticing health issues early since pets can't express discomfort.

I'd say staying consistent with their checkups of any kind

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trusting vets

Overall fitness / health, it may include balanced diet, and sufficient physical activity.

Some times he doesn't wana eat anything i have and just keep meowing all day long

Giving them medication.

Cleaning their mess

Limited access to vets

None

trimming nails

Finding a good and reliable vet

Spotting early signs of illness

Expenses

Travel and financing

cutting their nails

Food maybe

Difficult to notice when she's sick and keeping litter box clean

The biggest challenge is recognizing when something's off, since cats are really good at hiding pain or illness. It can be hard to tell if it's something minor or if a vet visit is needed.

Keeping a schedule of its food

Unable to identify the diagnosis

Not so much but keeping there cage clean 🧼🔪



How do you currently keep track of your pet's medical records or vaccination history?

28 responses

Manual methods like notebooks or digital tools like spreadsheets or notes or reminders on phones.

I write that in notes

Via clinics vaccination record book

Through a sheet given by vet

a notebook

I don't.

Get it checked every month

store my pet's medical records and vaccination history in a physical folder provided by the vet.

A booklet provided by the vet

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How do you currently keep track of your pet's medical records or vaccination history?

28 responses

through a pet passport

I didnt. Just had the basic vaccination done and nothing more was required.

I do not have any of his medical records

manually

A booklet.

The vet files

In my notes app

I only take my duck to the vet in case of any concern otherwise there are no medical records

through a booklet given by the vet

On paper

Vet issued booklet

Written records

A record book

idk

A file

We keep a record book provided by our vet for vaccinations and medical history

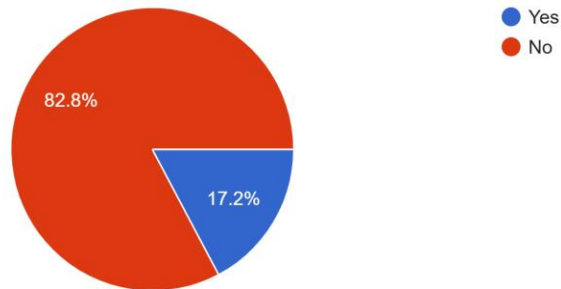
folder with all the vet receipts and records

In phone notes



Have you faced difficulties in scheduling vet appointments?

29 responses



If yes, what difficulties did you face?

7 responses

Getting the same time as stated and also making sure that it aligns with my schedule along with this it aligns with the needs of the cat.

Pet hospital is far away from my home takes lots of time

If we go on the appointed time we still have to wait for our turn for a long time

Vet didn't picked up the call

Scheduling

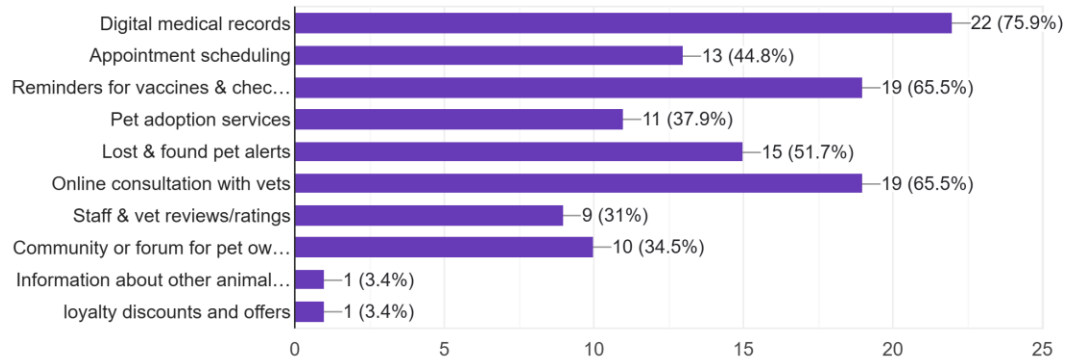
Nil

lack of trust, vet availability and expenses



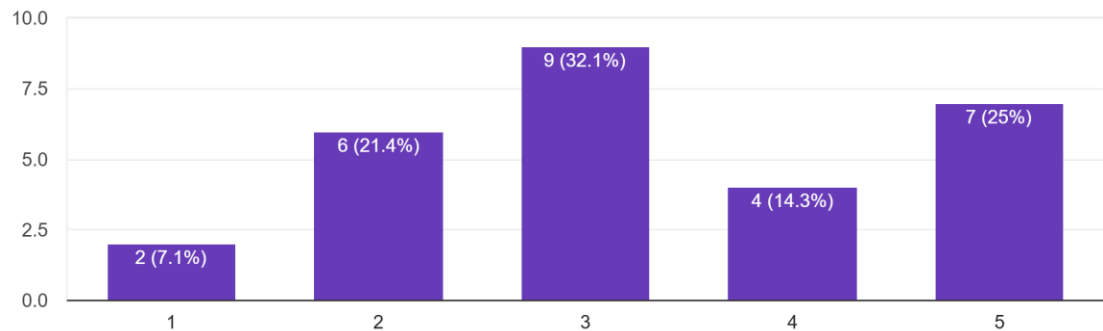
Which of these features would you find most useful in a pet app?

29 responses



How likely are you to use a pet care app regularly?

28 responses

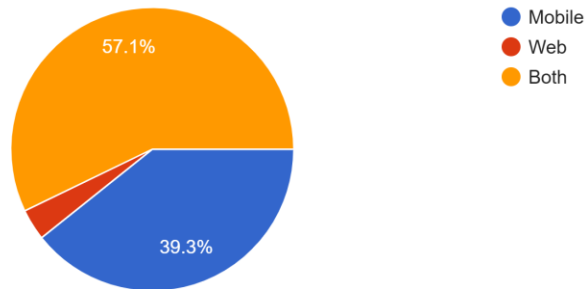


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Would you prefer the app to be mobile-only or both mobile & web?

28 responses



What would make you trust and continue using a pet care app?

22 responses

Ease of Use, Security, Reliability, Customization.

Consistency

Correct and timely data entry

By giving review

provides accurate reminders, secure record storage, and reliable vet recommendations.

User friendly interface

Credible sources and vets

Vaccines checkups and appointment customised according to my pet her age breed and health with the profile being able to update along with my pet

Some tips and care advices along with things that come with seasonal changes

And one not commonly known is the dental health of pets so that too

if it's reliable

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What would make you trust and continue using a pet care app?

22 responses

Trust would mean not having your location and mobile info tracked/recorded. For consistent use, regular notifications/popups or reminders, intuitive UI/UX, beneficial knowledge

yes

Dedication to the consumers.

Proper functioning and user friendly interface

Credibility

Ad free and no subscription required

a two way simultaneous response between owner and vet without delays

Availability and Consistency of the App Features

What would make you trust and continue using a pet care app?

22 responses

Availability and Consistency of the App Features

The credibility of vets and services, and ease of reach/access during emergencies in the system/app would make me (re)consider using a pet app.

Quality of services

Trusted vet recommendations and verified info.

I'd trust and keep using the app if it's easy to use, gives reliable info and sends helpful reminders for things like vet visits or medication

Its very convenient

Reminders about the vaccinations

If it have all the wet in my area and are not bombed with fake review will be good and a community to meet people is also a good inachative



- B. Assumption Logs

1. Staff & Roles:

Every member of staff is associated with a single role (Admin, Veterinarian, Volunteer).

Admins and Veterinarians are the only salaried employees, and all others are Volunteers.

2. Animal Intake & Shelter Rules:

Every animal should always be located in a certain shelter room.

There should be no surpassing of room capacity (system is supposed to control this).

Animal age is determined by either Date of Birth or it is estimated at the time of intake where Date of Birth is not known.

3. Adoption Process:

Only one animal can be adopted, after adoption, its status is changed into Adopted.

Adoption must be valid and adopters (CNIC, phone, address) are assumed to be genuine.

The adoption records are not the files that are stored but just links/paths to scanned documents.

4. Appointments & Treatments:

Veterinarians are the only people who can be appointed and treated.

An appointment (optional) does not have to be made in every animal.

All the treatments shall be associated to a single animal and a single vet.



5. Volunteers

One can assign several animals to a volunteer, and volunteers to animals.

The assumption is that volunteer shifts are recorded in blocks of a week (HoursPerWeek attribute).

5. System Operation & Data

The system will operate at regular times in the shelter (08:00-22:00).

Policy will be to make backups daily.

It is assumed that there is an Internet access that is used by the system.